π_0 · arXiv 2410.24164 · RSS 2025

Scientific Reviewer

8+3 · seat 4

Strengths · weaknesses · verdict
Systems+empirical RSS shift

Sep 28 paper club · VLA Architectures

Spine: PaliGemma + OT-CFM action chunks (DDPM→flow)

Physical Intelligence · H=50 · ~10 Euler · ~73 ms RTX4090 · NOT a classical planner

Club spine · one sentence

SigLIP — vision → **PaliGemma** — VLM → **π_0** — FM+chunks → **OpenPI** — release → **$\pi_{0.5}$/RTC** — async → **BEH'26** — defaults

π_0 popularized flow matching + action chunking on a VLM backbone

as the practical alternative to autoregressive discrete VLAs for dexterous 50 Hz control.

NOT a classical planner · generative continuous chunk controller · knobs: H · T_a · NFE · async · Σ

Strengths

1. Integrative	Architecture + data + systems + multi-robot eval as one framework
2. Right bias	FM+chunks vs AR VLAs; motivated by DP/ACT failure modes
3. Scale	~10k h + OXE; 7 embodiments; multi-minute tasks
4. Ablations	π_0-small · parity 160k · scratch vs FT · OOB vs FT
5. Honesty	Partial-credit rubrics; admit unreliability & opaque mixture

Weaknesses / threats to validity

Figure-only floats	Hard to meta-analyze without digitizing bars
π_0-small confounds	Size + init + DiT architecture all differ
Baseline compute	Parity helps; high-Hz data may still disadvantage OpenVLA/Octo
n=10 · little variance	Strong claims, thin statistical reporting
External validity	PI robots / home manip; not locomotion
“First FM VLA”	Treat as among first at this dexterity/scale

Verdict

Strong systems + empirical RSS paper

Correctly shifted the field’s default VLA action head

from AR tokens toward flow / diffusion chunks.

Scientific core holds · cite figures carefully for numeric claims

Private-data limits are real — do not oversell reproducibility

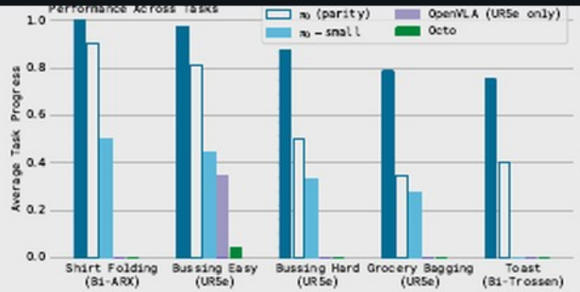


Fig. 7: **Out-of-box evaluation results**: We evaluate π_0 trained for the full 700k steps, a version trained for 160k steps that matches the number of updates for baseline models, π_0-small, and three baselines: OpenVLA and Octo trained on all of our data, and OpenVLA trained only on the UR5e tasks (which we found to work better on UR5e tasks). Across all tasks and all comparisons, even the "parity" version of our model outperforms all baselines, and the full version of our model achieves the best results by a large margin.

below the performance of π_0. Octo does support action chunks, but has a comparatively limited representational capacity. This comparison illustrates the importance of combining large, expressive architectures with the ability to model complex distributions via flow matching or diffusion. Additionally, the comparison to π_0-small illustrates the importance of incorporating VLM pre-training. Unfortunately, it is hard to make this last comparison fair: π_0-small uses fewer parameters, but larger models are difficult to use without pre-training. Overall, these experiments show that π_0 provides a powerful pre-trained model with the ability to effectively perform a variety of tasks with a variety of robots, with much better performance than prior models.

B. Following language commands

In the next set of experiments, we fine-tune the base π_0 model to follow language commands in a set of evaluation domains. We compare this fine-tuned π_0 model with the π_0-small model described in Section IV, which we found to be the strongest baseline in the previous section. Recall that π_0-small does *not* use a VLM initialization. This experiment therefore aims to measure how much VLM pre-training boosts our model's ability to follow language instructions.

Fig. 8: **The tasks in our language evaluation.** We evaluate our model on 3 different language-conditioned tasks, each of which requires following a sequence of intermediate language commands. The tasks involve bussing a table (top) to put dishes in a bin and garbage in a trash bin, setting a table (middle) by taking items out of a bin, and packing a shopping bag (bottom).

task consists of numerous such segments. The tasks in this evaluation consist of:

Bussing: the robot must clean a table, placing dishes and cutlery in a bin, and trash into a trash bin.

Table setting: the robot must take out items from a bin to set a table, including a place mat, dishes, silverware, napkin, and cups, and adjust them according to language instructions.

Grocery bagging: the robot must pack grocery items, such as bags of coffee beans, barley, marshmallow, seaweed, almonds, spaghetti, and cans into a bag.

In Figure 8, we show the language-conditioned tasks in our evaluation and present the evaluation results. We evaluate five different conditions. π_0-flat (and π_0-small-flat) corresponds to directly command the model with the task description (e.g., "bag the groceries"), without intermediate language commands. π_0-human (and π_0-small-human) provides intermediate step commands (e.g., which object to pick and where to place it) from an expert human user. These conditions evaluate each model's ability to follow more detailed language commands: while these intermediate commands provide considerable information for how to perform the task, the model must be able to understand and follow those commands to benefit from them. Finally, π_0-HL evaluates π_0 with high-level commands provided by a high-level VLM, as discussed in Section V-B. This condition is also autonomous, without any human expert.

The results in Figure 9, averaging over 10 trials per task, show that the language following accuracy of π_0 is significantly better than that of π_0-small. This suggests a significant improvement from the larger pre-trained VLM initialization. This capability translates to an improvement

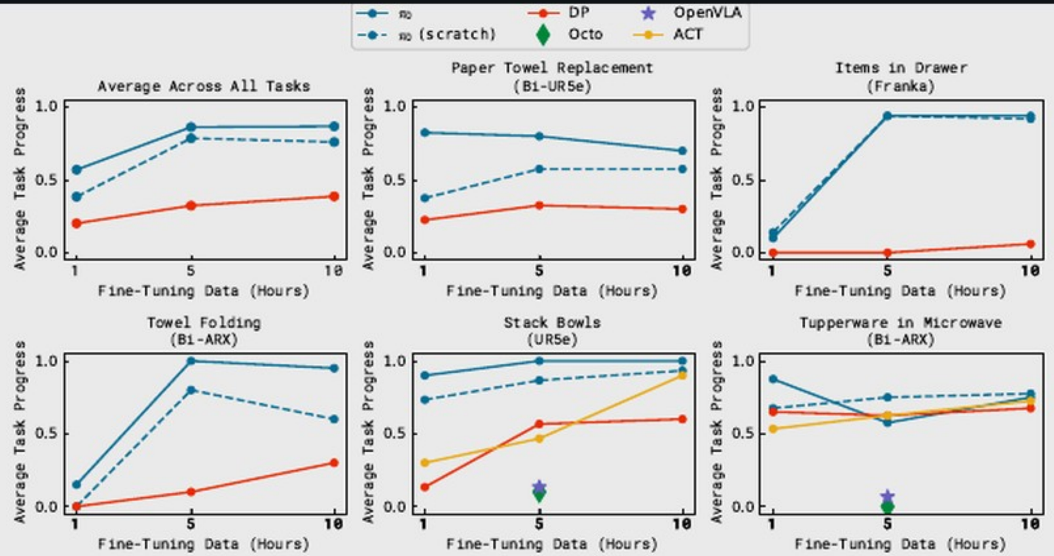


Fig. 11: **Fine-tuning with varying amounts of data.** π_0 can learn some easier tasks even with smaller amounts of data, and the pre-trained model often attains a larger improvement over the model trained from scratch.

that are more similar to the pre-training data, though the pre-trained model is frequently better than the non-pre-trained model, sometimes by as much as 2x.

D. Mastering complex multi-stage tasks

In our final set of experiments, we tackle a range of challenging multi-stage tasks via a combination of fine-tuning and language. For some of these tasks, data is present in pre-training, but fine-tuning is required to attain mastery. For some, no data is present in pre-training. The tasks in this evaluation, shown in Figure 12, are:

Laundry folding: This task requires a static (non-mobile) bi-manual system to fold articles of clothing. The clothing items start in a randomized crumpled state in a bin, and the goal is to take out the item, fold it, and place it on top of a stack of previously folded items. The randomized initial configuration of the crumpled laundry presents a major challenge, since the policy needs to generalize to any configuration. This task is present in pre-training.

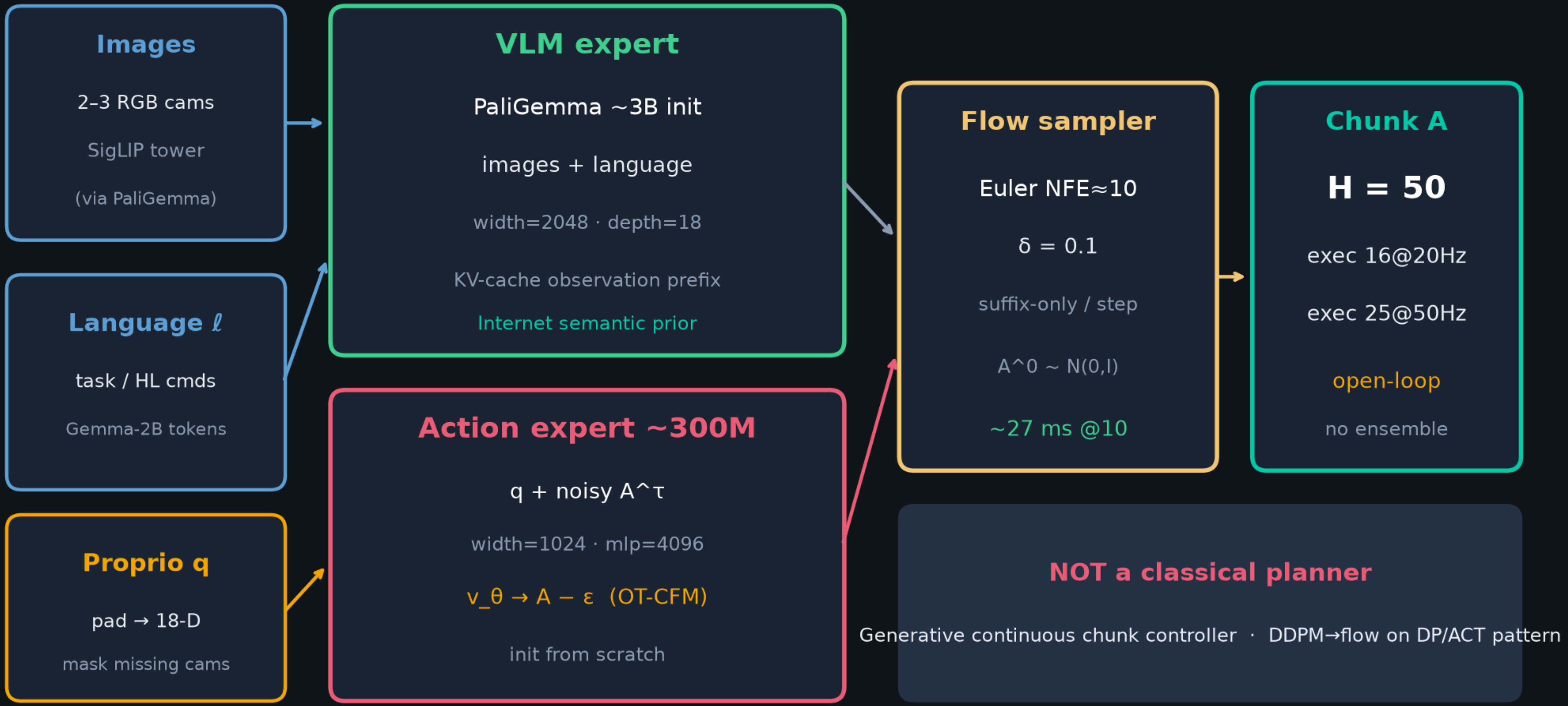
Mobile laundry: Here, the Fibocom mobile robot in Figure 5 has to fold laundry, facing many of the same challenges while of varying shapes and sizes, and perform complex dexterous motions, such as twisting the gripper to pick up large plates and carefully grasping thin, delicate items such as glasses. The robot must handle dense clutter and intelligently sequence various behaviors — for example, to clean off a plate with trash, it must first pick up the plate, then shake its contents into the garbage, and then place the plate in the bin. This task is not present in pre-training.

Box building: The robot has to assemble a cardboard box that starts in a flattened state. This task presents a number of major challenges: the box needs to bent in the right way, and the robot needs to hold down parts of the box while folding others, utilizing both arms and even the surface of the table to brace during folding motions. The robot might need to retry some folds, requiring a reactive and intelligent strategy. This task is not present in pre-training.

To-go box: This task requires moving several food items from a plate into a to-go box, requiring packing the items into the box so that they do not stick out, and then closing the box with both arms. This task is not present in pre-training.

Architecture · PaliGemma + OT-CFM action expert

~3.3B total · MoE dual-stream (Transfusion-inspired) · blockwise causal attention



Scientific Reviewer · close

Verdict: Strong systems+empirical RSS. Field shifted to flow/diffusion chunks.

Cite figures carefully · private-data limits honest

Q&A · Sep 28 · π_0 / 2410.24164

π_0: A Vision-Language-Action Flow Model for General Robot Control

Physical Intelligence
Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai,
Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine,
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